

Chapter 26 Current and Resistance

Electric Current

Current: the **directional movement** of **electric charges** forms a **electric current**.

Electric field is **set up** in the **speed of light**; electrons directional drift motion is 10^{-4}m/s ; random thermal motion is 10^6m/s .

Current and Current Density

Current definition

$$I = \frac{dQ}{dt}$$

Magnitude: The time rate of charge transfer through the **cross-sectional area** A .

Direction: The direction of directional movement of **positive charges**.

Current density:

$$\vec{J} = nq\vec{v}_d$$

Magnitude: current intensity per unit cross-sectional area at this point.

Direction: The direction of directional movement of the positive charge at this point.

n is moving charged particles per unit volume, q is the charge of single charged particle, $\vec{v}_d$ is drift speed.

And for current density and current:

$$I = \int \vec{J} \cdot d\vec{A}$$

Resistivity

Ohm's Law (Differential Form)

$$\vec{E} = \rho \vec{J}$$

where ρ is the **resistivity of material**. The reciprocal of resistivity is **conductivity** $\sigma = \frac{1}{\rho}$.

Ohm's Law holds true only for **linear materials under certain conditions**.

Resistivity changes with temperature:

$$\rho(T) = \rho_0[1 + \alpha(T - T_0)], \Delta T < 100\text{C}^\circ$$

where α is temperature coefficient of resistivity. For metal, $\alpha > 0$; for semiconductor $\alpha < 0$.

Resistance

Ohm's Law (Integral Form)

$$V = RI$$

And resistance

$$R = \int \frac{\rho}{A} dl$$

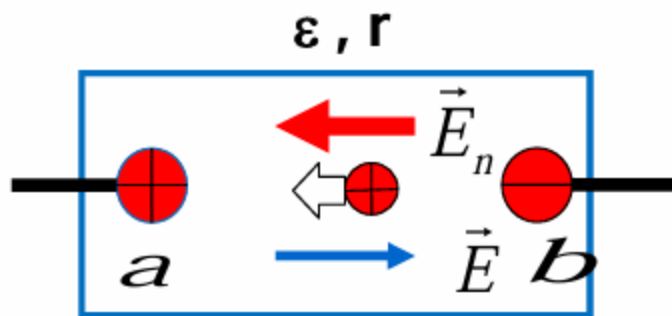
And resistance changes with temperature

$$R(T) = R_0[1 + \alpha(T - T_0)]$$

Electromotive Force and Circuit

Power source is a device that drive the positive charges from a low potential to a high potential and convert other forms of energy into electrical energy.

Electromotive force ϵ is equal to the **work done by the non-electrostatic force** that move a unit positive charge from negative to positive terminal.

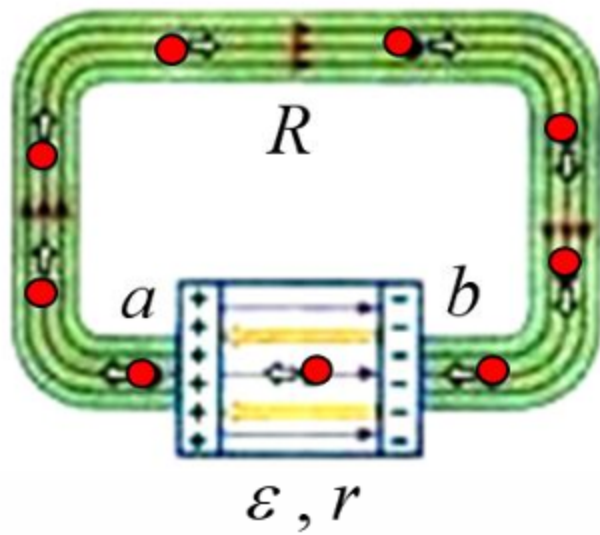


$$\epsilon = \int_{-}^{+} \vec{E}_n \cdot d\vec{l}$$

$$\epsilon = \int_{-}^{+} \vec{E}_n d\vec{l}$$

where $\vec{E}_n$ is the non-electric field strength.

Internal resistance r is the resistance of power source.



Closed-circuit:

$$V_{ab} = \mathcal{E} - Ir$$

$$V_{ab} = IR$$

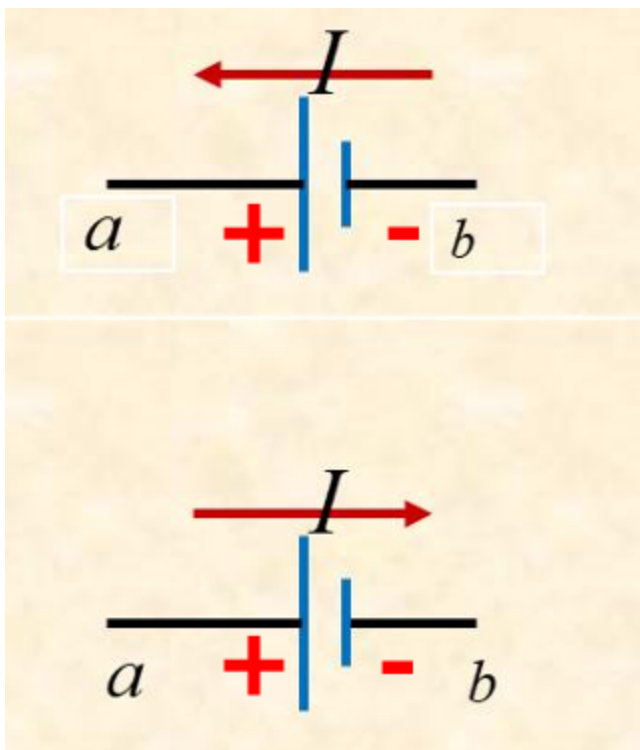
Open-circuit:

$$\mathcal{E} = V_{ab}$$

Ohm's Law for the entire circuit:

$$\mathcal{E} = I(R + r)$$

Energy and Power in Electric Circuits



Output Power:

$$P = V_{ab}I = \epsilon I - I^2 r$$

Input Power:

$$P = V_{ab}I = \epsilon I + I^2 r$$